



AMRITA
VISHWA VIDYAPEETHAM
AMRITAPURI CAMPUS
Amrita School of Engineering
B.Tech. Mid-Term Examination
Semester 5 – EAC (Academic Year 2026-2027)
23EAC302: Data Communication and Networks

Duration – 2 Hours

Maximum Marks – 50

Instruction: Answer all questions.

Roll No. AM:EN:24EAC24035....

1. Short Answer Questions. [Each question carries 2 marks]
- The Ethernet protocol operates at the Tcp and Udp layer of the OSI model.
 - You have an interface on a router with the IP address of 201.188.110.10/29. The network address and broadcast address used by the hosts on this LAN are 201.188.110.0 and 201.188.110.7.
 - For a small network, classless interdomain routing is used as the IP assignment scheme, what is the optimum IP address mask 'n' in the IP format "x.y.z.w/n" to be used on a point-to-point link to reduce the wastage of IP addresses? Justify your answer.
 - What is the primary difference between flow control applied at the data link layer and transport layer?
 - A computer network consists of 18 devices connected using a full mesh topology. Calculate the total number of point-to-point links required to connect all the devices.
 - Explain how an IPv6 address is different from an IPv4 address. Can an IP address of one format be converted to another? [12 Marks] [CO 1] [L2]
2. For a communication link between two end nodes A and B, there are two intermediate nodes X and Y. For the transmission of a string of packets from A to B, state the highest layer of the OSI model that will be operating at each node. Briefly justify your claim. [4 Marks] [CO 1] [L2]
3. The ECE department is planning to establish a small Digital Experience Centre within the department library. The centre will be furnished with systems for the following sections:
- Students' Corner: 30 computers used by students
 - Faculty Corner: 10 computers used by faculty members
 - Admin purposes: 5 computers used by staff
- The department wants the network to be reliable, easy to manage, and scalable so that additional computers can be added in the future. As a network administrator:
- Recommend a suitable network topology for the Digital Experience Centre and justify your choice. (3 Marks)
 - Draw a neat network diagram showing the computers, networking devices, and the interconnections and all the intermediate devices involved. (3 Marks)
 - State two advantages and a disadvantage of the topology selected. (3 Marks)
- [9 Marks] [CO 1] [L2]

4. You have decided to set up a café with a gaming arcade in Vallikavu, seeing the immense potential for such a business. You are in the process of setting up a new network infrastructure to support its café operations, game development, customer support, delivery team and finance. The company has been assigned the IPv4 network 172.16.50.0/24. Answer the following questions regarding the design of your network deployment.

Section	Number of hosts
Café operations	25
Game development	45
Customer support	12
Delivery team	60
Finance	6

- Determine the appropriate prefix length for each department. (5 Marks)
- Find the subnet masks of all departments, starting with the largest host requirement. (5 Marks)
- For each subnet, determine the following: (4 Marks)
 - Network address
 - Range of host addresses
 - Broadcast address
- Calculate the number of unused IP addresses remaining in the original /24 network after all five subnets have been allocated. (1 Mark)

[15 Marks] [CO 2] [L3]

5. An employee is unable to access the company's internal web application from their desktop computer. The network administrator investigates the problem and observes the following:
- The computer is powered on and the network cable is properly connected.
 - The network interface shows that the computer has a valid **MAC address**.
 - The computer has been assigned an **IP address**, but it cannot communicate with the application server.
 - After correcting the routing configuration, the computer can reach the server, but the web application still does not load.

Answer the following questions for the above scenario:

- Identify the OSI layer(s) associated with each of the problems described above.
- Explain the role of the relevant OSI layer in each case.
- Arrange the OSI layers involved in establishing communication between the employee's computer and the application server in the correct order.

[4+4+2 = 10 Marks] [CO 2] [L3]

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